

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1497

James De Young 230 kV
September 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1497. The Interconnection Customer proposes a 98.0 MW battery energy storage facility interconnecting at the James De Young 230 kV substation in the Ohio area. The request was received on June 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 511 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,763,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

MARBLE - SOLSBERRY 100kV: 145.5% loading (136 MVA) following the outage of the BURRIS PIT - JAMES DE YOUNG 230kV; DFAX = 5.6%.

The MASON CRK - CARLEY ROAD 100kV is overloaded to 141.1% of its 111 MVA emergency rating for the loss of the CARLEY RD - URANA SUBSTATION 100kV. The project contributes 5.7% DFAX (5.6 MW) to this overload.

HENDRON - MARBLE 100kV: 137.0% loading (132 MVA) following the outage of the BURRIS PIT - JAMES DE YOUNG SUBSTATION 230kV Line; DFAX = 5.6%.

For the loss of JAMES DE YOUNG - BURRIS PIT 230kV Line, the HENDRON - CAMBRIDGE 100kV Line loads to 134.7% of its 128 MVA rating. Project DFAX: 5.6%.

CARLEY RD - URBANA 100kV: 133.9% loading (123 MVA) following the outage of the CARLEY ROAD - MASON CREEK SUBSTATION 100kV line; DFAX = 6.1%.

The MECHANICSBURG - CAMBRIDGE 100kV is overloaded to 132.0% of its 125 MVA emergency rating for the loss of the BURRIS PIT SUBSTATION - JAMES D YOUNG 230kV. The project contributes 5.6% DFAX (5.5 MW) to this overload.

The BARRETT SUBSTATION - MINCO 230kV is overloaded to 112.0% of its 723 MVA emergency rating for the loss of the BARRETT - NORWICH 500kV. The project contributes 5.2% DFAX (5.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MARBLE - SOLSBERRY SUBSTATION 100kV	Rebuild line with higher-capacity conductor	\$4,132,000	100% (\$4,132,000)
MASON CRK - CARLEY RD 100kV Line	Rebuild line with higher-capacity conductor	\$1,294,000	100% (\$1,294,000)
HENDRON - MARBLE 100kV	Rebuild line with higher-capacity conductor	\$8,540,000	100% (\$8,540,000)
HENDRON SUBSTATION - CAMBRIDGE SUBSTATION 100kV	Rebuild line with higher-capacity conductor	\$4,157,000	100% (\$4,157,000)
CARLEY RD SUBSTATION - URBANA 100kV	Rebuild line with higher-capacity conductor	\$2,987,000	100% (\$2,987,000)
MECHANICSBURG - CAMBRIDGE 100kV	Rebuild line with higher-capacity conductor	\$3,476,000	100% (\$3,476,000)
MINCO - BARRETT 230kV	Rebuild line with higher-capacity conductor	\$14,577,000	1% (\$177,000)
	Total allocated to project		\$24,763,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.